
Algorithm 1 Subgradient Algorithm for MAX-APC

Input: Bipartite graph $G = (V_1 \cup V_2, E)$, profits p_e for $e \in E$, conflict graph $\delta_c = (V(\delta_c), E(\delta_c))$, max iterations $K_{\max}$, tolerance ε

Output: Best feasible lower bound LB , upper bound UB , best feasible solution x^{LB}

— **Initialization** —

- 1: **Instance Generation:** Construct the full bipartite instance with $|E| = n^2$ edges. The first n edges $E^0 \subseteq E$ form a valid complete assignment with *no conflicts*. The remaining $n^2 - n$ edges receive weights and conflict pairs, defining $\delta_c = (V(\delta_c), E(\delta_c))$.
- 2: $z^{(0)} \leftarrow \sum_{e \in E^0} p_e$
- 3: $LB \leftarrow z^{(0)}$, $x^{LB} \leftarrow \mathbf{1}_{E^0}$ ▷ Best conflict-free solution; never directly modified
- 4: Solve relaxation over all n^2 edges without conflict constraints via Hungarian $\Rightarrow z_{\text{relax}}^{(0)}$
- 5: $UB \leftarrow z_{\text{relax}}^{(0)}$ ▷ Initial UB from the unconstrained relaxation
- 6: $k \leftarrow 0$, $t^{(0)} \leftarrow 0$, $\pi^{(0)} \leftarrow 2$, $\lambda_{ef}^{(0)} \leftarrow 0 \quad \forall \{e, f\} \in E(\delta_c)$

— **Main Loop** —

- 7: **repeat**
- 8: $k \leftarrow k + 1$
- 9: *Step 1 – Penalize conflicting edges and solve*
 $\tilde{p}_e^{(k)} \leftarrow p_e - \sum_{\{e, f\} \in E(\delta_c)} \lambda_{ef}^{(k-1)} \quad \forall e \in E$
- 10: Solve Max. Weight Assignment with $\tilde{p}^{(k)}$ via Hungarian $\Rightarrow x^*$, $z_1^*(\lambda^{(k-1)})$
 Step 2 – Lagrangian upper bound (unconditional)
- 11: Compute and **unconditionally** set:

$$UB \leftarrow Z_{\text{Lag}} = z_1^*(\lambda^{(k-1)}) + \sum_{\{e, f\} \in E(\delta_c)} \lambda_{ef}^{(k-1)}$$

Step 3 – Feasibility, optimality, and LB update

- 12: **if** $x_e^* + x_f^* \leq 1$ for all $\{e, f\} \in E(\delta_c)$ **then** ▷ Conflict-free
 - 13: **if** $x_e^* + x_f^* = 1$ for all $\{e, f\} \in E(\delta_c)$ with $\lambda_{ef}^{(k-1)} > 0$ **then**
 - 14: **STOP** — Complementary Slackness satisfied; **return** x^* , Z_{Lag}
 - 15: **end if**
 - 16: **if** $\sum_{e \in E} p_e x_e^* > LB$ **then**
 - 17: $LB \leftarrow \sum_{e \in E} p_e x_e^*$, $x^{LB} \leftarrow x^*$, $t^{(k)} \leftarrow 0$
 - 18: **else**
 - 19: $t^{(k)} \leftarrow t^{(k-1)} + 1$
 - 20: **end if**
 - 21: **else** ▷ Has conflicts — repair; x^{LB} never directly modified
 - 22: Apply **Repair Heuristic** to $x^* \Rightarrow$ feasible $\hat{x}$, $\hat{z}$
 - 23: **if** $\hat{z} > LB$ **then**
 - 24: $LB \leftarrow \hat{z}$, $x^{LB} \leftarrow \hat{x}$, $t^{(k)} \leftarrow 0$
 - 25: **else**
 - 26: $t^{(k)} \leftarrow t^{(k-1)} + 1$
 - 27: **end if**
 - 28: **end if**
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Algorithm 1 Subgradient Algorithm for MAX-APC (continued)

Step 4 – Step-size halving

29: **if** $t^{(k)} \geq 20$ **then**

30: $t^{(k)} \leftarrow 0, \quad \pi^{(k)} \leftarrow \pi^{(k-1)}/2$

31: **else**

32: $\pi^{(k)} \leftarrow \pi^{(k-1)}$

33: **end if**

Step 5 – Subgradient and multiplier update

34:

$$s_{ef}^{(k)} = 1 - x_e^* - x_f^* \quad \forall \{e, f\} \in E(\delta_c)$$

$$\|s^{(k)}\|^2 = \sum_{\{e,f\} \in E(\delta_c)} [s_{ef}^{(k)}]^2$$

35:

$$\alpha^{(k)} = \pi^{(k)} \cdot \frac{Z_{\text{Lag}} - LB}{\|s^{(k)}\|^2}$$

36:

$$\lambda_{ef}^{(k)} = \max\left\{0, \lambda_{ef}^{(k-1)} + \alpha^{(k)}(1 - x_e^* - x_f^*)\right\} \quad \forall \{e, f\} \in E(\delta_c)$$

37: $k = K_{\max}$ or $\pi^{(k)} < \varepsilon$

38: **Report** UB, LB, x^{LB} and **end.** =0
